

Deterministic geometric Savitzky-Golay dynamics estimator

Shared parameter update: reference -> estimated

mass [kg]	2.351597591 -> 2.351597591	delta=+4.38736e-10	ratio=1
CoG x [m]	-0.002024708562 -> -0.002024708568	delta=-5.47911e-12	
ratio=-			
CoG y [m]	-3.052657894e-05 -> -3.052655112e-05	delta=+2.78241e-11	
ratio=-			
CoG z [m]	0.009509749599 -> 0.009509749616	delta=+1.66588e-11	
ratio=-			
rotor effectiveness 1	1 -> 0.9999999993	delta=-7.2827e-10	ratio=1
rotor effectiveness 2	1 -> 0.9999999994	delta=-6.09293e-10	ratio=1
rotor effectiveness 3	1 -> 0.9999999999	delta=-1.00035e-09	ratio=1
rotor effectiveness 4	1 -> 0.9999999989	delta=-1.06708e-09	ratio=1
command lag [s]	0 -> 0.004000000392	delta=+0.004	ratio=-

Inertia matrix [kg m²]

reference:

[0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[1.901508003e-05, 5.9167305e-08, 0.1289921107]

estimated:

[0.06500006149, -7.279002842e-07, 1.90150801e-05]
[-7.279002842e-07, 0.06495265636, 5.916787922e-08]
[1.90150801e-05, 5.916787922e-08, 0.1289921107]

Principal moments: reference -> estimated

principal moment 1 [kg m ²]	0.06495264516 -> 0.06495264518	delta=+1.56455e-11	
ratio=1			
principal moment 2 [kg m ²]	0.06500006701 -> 0.06500006702	delta=+8.41772e-12	
ratio=1			
principal moment 3 [kg m ²]	0.1289921163 -> 0.1289921163	delta=+8.08281e-12	ratio=1

Optimization

joint dynamics loss: 2.81951540316

Gaussian-prior cost: 2.036129923e-16

selected strict-ZOH lag [s]: 0.004000000392

Bag failure_1_single

reconstruction position RMSE [m]: 0.000497212640177

reconstruction position component RMSE [m]: [0.000302 0.000297 0.000261]

reconstruction orientation RMSE [deg]: 0.528821719301

gyro consistency RMSE xyz [rad/s]: [0.060846 0.087243 0.042587]

specific-force consistency RMSE xyz [m/s²]: [0.906473 0.204361 0.366322]

inferred external force RMS xyz [N]: [2.033545 0.63799 4.088639]

inferred external torque RMS xyz [N m]: [0.154586 0.551072 0.217628]

bag dynamics loss: 2.81951540316

Recorded command timestamp intervals (data-derived)

Bag failure_1_single

rotor: median dt=0.00499773026 s (200.091 Hz), min=2.00271606e-05 s, max=0.0138893127 s

gimbal: median dt=0.00499868393 s (200.053 Hz), min=4.529953e-06 s, max=2.48282003 s